



**MBERE ML: A Machine Learning Model for Driver-Level  
Road-Accident-Risk Prediction in the Rwandan Motor-Insurance and  
Fleet Context**

**BSc. in Software Engineering**  
**MANZI Terry**

Supervisor: **Emmanuel Adjei**

Date: [Submission date]

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## **List of Acronyms / Abbreviations**

**API** Application Programming Interface

**AUC** Area Under the (ROC) Curve

**BNR** National Bank of Rwanda (Banque Nationale du Rwanda)

**ERD** Entity-Relationship Diagram

**GDP** Gross Domestic Product

**GPS** Global Positioning System

**LMIC** Low- and Middle-Income Country

**ML** Machine Learning

**RF** Random Forest

**RNP** Rwanda National Police

**RTA** Road Traffic Accident

**RTC** Road Traffic Crash

**SHAP** SHapley Additive exPlanations

**SMOTE** Synthetic Minority Over-sampling Technique

**SVM** Support Vector Machine

**UML** Unified Modeling Language

**WHO** World Health Organization

**XAI** Explainable Artificial Intelligence

**XGBoost** Extreme Gradient Boosting

# CHAPTER ONE: INTRODUCTION

## 1.1 Introduction and Background

Road traffic crashes are a major and growing global public health problem. According to the World Health Organization (WHO, 2023a), approximately 1.19 million people die on the world's roads every year, and road traffic injuries are the leading cause of death for children and young adults aged 5 to 29 years. The burden is highly unequal: about 92% of road fatalities occur in low- and middle-income countries (LMICs), even though these countries hold only around 60% of the world's vehicles, and the WHO African Region records the highest road traffic death rate of any region (WHO, 2023a). These deaths and injuries also carry a heavy economic cost; the World Bank (2018) estimates that LMICs could raise income per capita substantially by as much as 7 to 22% over a 24-year horizon in the countries it studied if they decreased road traffic deaths and injuries, underscoring that road safety is also an economic development issue.

Rwanda reflects this regional pattern. National records show a rising trend in both crashes and deaths: 4,203 crashes with 687 deaths in 2020, 8,639 crashes with 655 deaths in 2021, and 10,334 crashes with 729 deaths in 2022 (Bahati & Masabo, 2025), with the upward trend continuing in 2023 (IGIHE, 2025). Within Rwanda, crashes are heavily concentrated in Kigali, where Patel et al. (2016) mapped distinct road traffic injury hotspots from police data and found that victims were predominantly young males (mean age 35.9 years) and that motorcycles carried an elevated risk of severe injury. Motorcycle taxis are a dominant transport mode in Kigali, which shapes both the nature of crashes and the design of any locally relevant safety solution.

Much road-safety analysis treats the crash as the unit of study where crashes happen, how many occur, and how severe they are. An equally important but less-studied question is which individual drivers are most likely to be involved in a crash. Insurers and fleet operators, in particular, carry risk at the level of the individual driver, yet in Rwanda they lack data-driven tools to estimate that risk. This project, MBERE ML, addresses that gap by developing a machine-learning model that predicts driver-level accident risk from driver, vehicle and environmental features, contextualized for the Rwandan road environment, and delivers the result through an accessible platform for insurers and fleet operators.

## 1.2 Problem statement

First, road traffic crashes in Rwanda are frequent, rising and costly. National figures show crashes increasing from 4,203 in 2020 to 10,334 in 2022 (Bahati & Masabo, 2025), and road crashes impose large economic costs on developing economies, removing working-age adults from the labour force and consuming health and productivity resources (World Bank, 2018). Reducing crashes therefore has both a humanitarian and an economic justification, but doing so requires the ability to identify where risk is concentrated.

Second, the institutions that bear the financial consequences of crashes like motor insurers and fleet operators cannot currently estimate the accident risk of an individual driver. Rwandan insurers price motor cover largely through broad risk pooling rather than individual driver risk, so cautious drivers effectively subsidize risky ones. Motor insurance has consequently been a persistent loss-maker; motorcycle premiums have continued to rise, and in 2026 the regulator launched a National Insurance Strategy to address challenges such as slow claims and low penetration (The New Times, 2026). A driver-level risk estimate is the proposed input that would allow fairer, risk-based pricing and targeted safety intervention.

Third, there is a clear research gap. Machine learning has been applied to Rwandan road safety, but only at an aggregate level: Gatera et al. (2023) forecast short-term accident counts from national police data using Random Forest and Support Vector Machine models, and Bahati & Masabo (2025) used Random Forest and clustering to optimize ambulance placement and predict emergency response time. Neither predicts the risk of an individual driver. More broadly, machine-learning road-safety models for LMICs remain constrained by data scarcity and by the use of variables drawn from high-income settings that omit locally important factors such as unpaved roads, weather and a motorcycle-heavy fleet (Assaye, 2025).

To address these problems, this project proposes MBERE ML: a machine-learning model that predicts individual driver/road-user accident risk from driver, vehicle and environmental features. The model is trained and benchmarked on real East-African crash data, contextualized for Rwanda's road environment and motorcycle-dominated fleet, made interpretable through explainable-AI techniques, and delivered through a web platform on which motor insurers and fleet operators can view and analyse driver risk scores.

## **1.3 Main Objective**

The main objective of this project is to develop and evaluate a machine-learning model that predicts individual driver accident risk from driver, vehicle and environmental features contextualized for Rwanda and to deliver that capability through an accessible platform for motor insurers and fleet operators.

### **1.3.1 Specific Objectives**

1. To build and preprocess a driver-and-context accident dataset from real East-African police records (the Addis Ababa Road Traffic Accident dataset), selecting and engineering features relevant to Rwanda's road environment, and to request a Rwandan sample for contextual validation.
2. To develop and compare risk-prediction models a transparent rule-based baseline against machine-learning models (Random Forest and XGBoost) while handling class imbalance, and to evaluate them using precision, recall, F1-score and AUC.
3. To make predictions interpretable using explainable-AI techniques (SHAP) and to deliver them through a web platform on which insurers and fleet operators can view, filter and analyse driver risk scores.

## **1.4 Research Questions**

- RQ1. Which driver, vehicle and environmental features most strongly predict accident risk in an East-African LMIC road context relevant to Rwanda?
- RQ2. Do machine-learning models (Random Forest, XGBoost) outperform a rule-based baseline for driver-level accident-risk prediction, and which model performs best under class imbalance?
- RQ3. How can the model's predictions be made interpretable and actionable for Rwandan motor insurers and fleet operators?

## **1.5 Project Scope**

In scope, the project delivers a data-driven, driver-level accident-risk model. It uses the Addis Ababa Road Traffic Accident dataset as the primary training and evaluation source, with Rwanda-relevant feature selection and a request to the Rwanda National Police and the Rwanda Biomedical Centre for a local validation sample. It compares a rule-based baseline against Random Forest and XGBoost models, applies the Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance, uses SHAP for explainability, and provides a web-platform prototype for the identified beneficiaries motor insurers and fleet or motorcycle-taxi operators.

Out of scope, the project does not include on-vehicle sensors or real-time crash detection, nationwide deployment, or production integration with insurers' core policy and claims systems. Because driver-level telematics linked to claims are proprietary and not publicly available, the model predicts risk from the driver, vehicle and environmental attributes recorded in crash data rather than from continuous on-board telematics; collecting primary telematics data is left as future work. The platform is a functional prototype intended to demonstrate value, not a production system.

## **1.6 Significance and Justification**

Reducing road crashes carries a strong economic as well as humanitarian case: the World Bank (2018) estimates that substantial gains in income per capita are possible in LMICs that cut road deaths and injuries, so any tool that helps target prevention has measurable value. MBERE ML contributes a means of identifying higher-risk drivers so that limited safety resources can be focused where they matter most.

For the motor-insurance and fleet sectors, the contribution is direct. Motor insurance in Rwanda has been a persistent loss-maker priced largely through broad pooling, with rising premiums and a recent national reform effort (The New Times, 2026). A driver-level risk model supplies a valuable input for fairer, risk-based pricing and for targeted interventions such as driver feedback and training, benefiting insurers, fleet operators and lower-risk drivers alike.

Academically, the project offers a driver-level, behaviour-and-profile-based accident-risk model contextualized for Rwanda, moving beyond the aggregate forecasting, hotspot and severity work that characterizes existing Rwandan research (Gatera et al., 2023; Bahati & Masabo, 2025) and validating its approach on real East-African data rather than imported high-income datasets (Assaye, 2025).

## 1.7 Research Budget

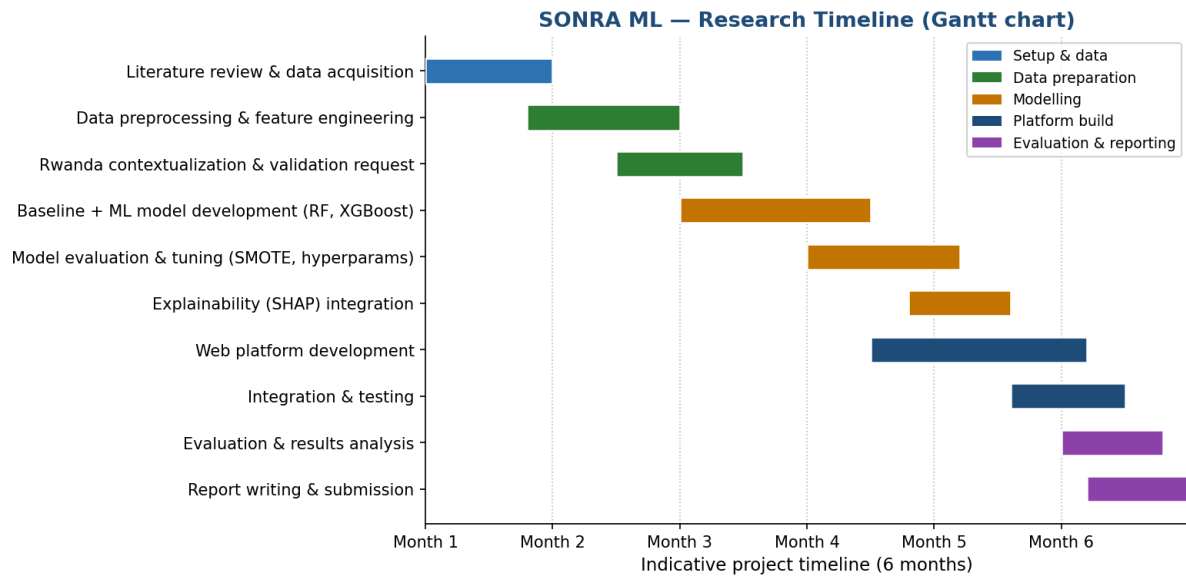
Because the project is only software and data-based, its cost is low. The primary datasets are open and free to access, model training can run on local machines or free cloud tiers, and only modest hosting and connectivity costs are anticipated. All figures below are estimates in United States Dollars (USD) and may vary with the provider and exchange rate.

*Table 1: Research budget (estimated project costs).*

| Item                                          | Qty | Unit cost (USD) | Subtotal (USD) |
|-----------------------------------------------|-----|-----------------|----------------|
| Open datasets (Addis Ababa RTA; Porto Seguro) | –   | 0               | 0              |
| Cloud compute (Colab Pro, optional, 2 months) | 2   | 10              | 20             |
| Platform hosting (prototype, ~6 months)       | 6   | 5               | 30             |
| Internet / data bundles and miscellaneous     | –   | –               | 10             |
| <b>Estimated total</b>                        |     |                 | <b>≈ 60</b>    |

## 1.8 Research Timeline

The Gantt chart in Figure 5 presents an indicative six-month schedule for the project, from the literature review and data acquisition through data preparation, model development, platform build, evaluation and final reporting. The phases overlap deliberately so that modelling and platform work can proceed in parallel once the data are prepared.



*Figure 5: Research timeline (Gantt chart).*

## **CHAPTER TWO: LITERATURE REVIEW**

### **2.1 Introduction**

This review targets software and machine-learning-related literature on driver-level accident-risk prediction, crash-severity and crash-frequency modelling, and the contextualization of such models for low and middle-income countries, with particular attention to work relevant to Rwanda and East Africa. Literature was identified through a keyword search of indexed academic platforms including Google Scholar, Semantic Scholar, PubMed Central, IEEE Xplore and reputable open-access publishers (for example MDPI, BMC, Springer Nature and Elsevier), supplemented by authoritative institutional sources such as the WHO and the World Bank. Search phrases included “driver accident risk prediction machine learning,” “vehicle crash severity prediction random forest,” “road accident prediction Rwanda,” “machine learning road safety low-income country,” and “usage-based insurance driver risk.” Sources were screened for relevance, recency and methodological quality, and a core set of peer-reviewed studies, official reports and datasets was retained for detailed review.

### **2.2 Overview of Existing Approaches**

Existing data-driven approaches to road-accident risk fall into a few broad categories. The first models risk at the level of place and time forecasting how many crashes will occur, or where hotspots lie typically from historical police records. The second predicts the severity of a recorded crash from contextual and driver attributes. The third, used mainly by insurers, predicts risk at the level of the individual driver, either from declared attributes or from telematics data that capture driving behaviour such as speed, braking and time of day. Each category answers a different question, and the data each requires differs accordingly: aggregate forecasting and severity models use crash-record data, whereas true driver-level behavioural risk prediction depends on driver-linked behaviour and claims data, which are largely held privately by insurers and telematics providers and are rarely available for an LMIC such as Rwanda.

This data reality shapes the present project. Because driver-level telematics linked to outcomes are not publicly available for Rwanda, the model is built on real crash-record data with interpretable, driver-and-context features, contextualized for the Rwandan environment, and

benchmarked against a publicly available driver-level insurance dataset. This positions the work between the severity-modelling and driver-risk traditions while remaining feasible within the constraints my project.

## **2.3 Review of Related Work**

In the Rwandan context, Gatera et al. (2023) compared Random Forest and Support Vector Machine models for short-term forecasting of road-accident counts using national police data, with the Random Forest performing best. Bahati & Masabo (2025) applied Random Forest together with clustering and optimization to Rwandan accident and emergency-response data to predict response time and optimize ambulance placement. Both studies demonstrate that machine learning is feasible and locally relevant, but both operate at the aggregate level counts, hotspots and logistics rather than predicting the risk of an individual driver.

On severity and risk-factor modelling, Shaffiee Haghshenas et al. (2025) compared machine-learning models with a logit baseline and showed that machine learning can classify the level of road-crash severity from real crash records. Most directly relevant to this project, Assaye (2025) predicted car-accident severity in Northwest Ethiopia using ten machine-learning models on 2,000 police-reported accidents, integrating driver demographics, behavioural factors and environmental conditions; Random Forest performed best (about 82% accuracy, AUC 0.87), the Synthetic Minority Over-sampling Technique (SMOTE) improved accuracy by addressing class imbalance, and SHAP provided interpretability. That study is a close methodological and regional precedent and informs the pipeline adopted here, while differing in that it predicts crash severity rather than driver-level risk.

On the insurance side, the publicly available Porto Seguro (2017) Safe Driver Prediction dataset frames the exact problem of interest predicting the probability that an individual driver will file an auto-insurance claim and motivates risk-based pricing, although its features are anonymized and it derives from a Brazilian rather than a Rwandan context. The primary training dataset for this project, the Addis Ababa Road Traffic Accident dataset compiled by Bedane (2020) from sub-city police records, provides real East-African data with interpretable driver, vehicle, road and environmental features suited to the Rwanda-relevant feature analysis proposed here.

### 2.3.1 Summary of Reviewed Literature

Table 2: Summary of reviewed literature.

| Author / Source            | Year | Method                                              | Key findings                                                                                   | Relevance to MBERE ML                                                                |
|----------------------------|------|-----------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Gatera et al.              | 2023 | Random Forest & SVM forecasting on RNP data         | Random Forest forecast short-term accident counts best.                                        | Closest Rwandan prior work; aggregate counts, not driver-level risk.                 |
| Bahati & Masabo            | 2025 | Random Forest + clustering + optimization           | Predicted response time and optimized ambulance hotspots; rising national crash figures.       | Local ML feasibility; post-crash logistics, not pre-crash driver risk.               |
| Assaye                     | 2025 | 10 ML models + SMOTE + SHAP on Addis police data    | Random Forest best (~82% acc, AUC 0.87); SMOTE improved imbalance; SHAP gave interpretability. | Direct East-African LMIC precedent; informs the pipeline; severity, not driver risk. |
| Shaffiee Haghshenas et al. | 2025 | ML models vs. logit baseline                        | ML classifies road-crash severity from real crash records.                                     | Supports ML over baseline for crash-risk modelling.                                  |
| Porto Seguro               | 2017 | Driver-level claim-probability prediction (insurer) | Models predict the probability a driver files a claim; enables risk-based pricing.             | Public driver-level benchmark matching the insurer use case.                         |
| Bedane                     | 2020 | Police-record dataset (Addis Ababa)                 | 12,316 crash records with interpretable driver, road and environment features.                 | Primary training dataset for this study.                                             |

### 2.4 Strengths and Weaknesses of Existing Approaches

Table 3: Strengths and weaknesses of existing approaches.

| Approach                                                                 | Strengths                                                                            | Weaknesses / gaps for Rwanda                                             |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Aggregate crash forecasting (e.g., Gatera et al., 2023)                  | Locally validated; useful for planning and resource allocation.                      | Predicts counts and hotspots, not which drivers are at risk.             |
| Crash-severity ML (e.g., Assaye, 2025; Shaffiee Haghshenas et al., 2025) | Accurate; uses interpretable driver and environment features; LMIC precedent exists. | Predicts severity of recorded crashes, not prospective driver risk.      |
| Telematics / usage-based insurance models                                | Strong driver-level behavioural risk prediction.                                     | Require proprietary telematics linked to claims; no public Rwandan data. |

| Approach                                                | Strengths                                                      | Weaknesses / gaps for Rwanda                                                 |
|---------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------|
| Public driver-level claims data<br>(Porto Seguro, 2017) | Genuine driver-level risk target;<br>matches insurer use case. | Anonymized features;<br>non-Rwandan; no local<br>contextualization possible. |
| Ambulance-optimization ML<br>(Bahati & Masabo, 2025)    | Improves emergency-response<br>logistics in Rwanda.            | Post-crash focus; does not estimate<br>driver risk before a crash.           |

## 2.5 General Comment and Conclusion

The reviewed literature confirms that the components of MBERE ML are feasible and that the chosen direction is well grounded. Machine learning is already applied to Rwandan road safety, but only at the aggregate level of counts, hotspots and logistics (Gatera et al., 2023; Bahati & Masabo, 2025). Crash-severity modelling with interpretable features and class-imbalance handling has been demonstrated on real East-African police data (Assaye, 2025), and the insurance sector's own framing of driver-level risk is captured in public benchmarks (Porto Seguro, 2017). The gap is that no published Rwandan study predicts driver-level accident risk using locally relevant features. MBERE ML is positioned to fill exactly this gap, building on the Addis Ababa dataset (Bedane, 2020) and contextualizing it for Rwanda's motorcycle-dominated road environment.

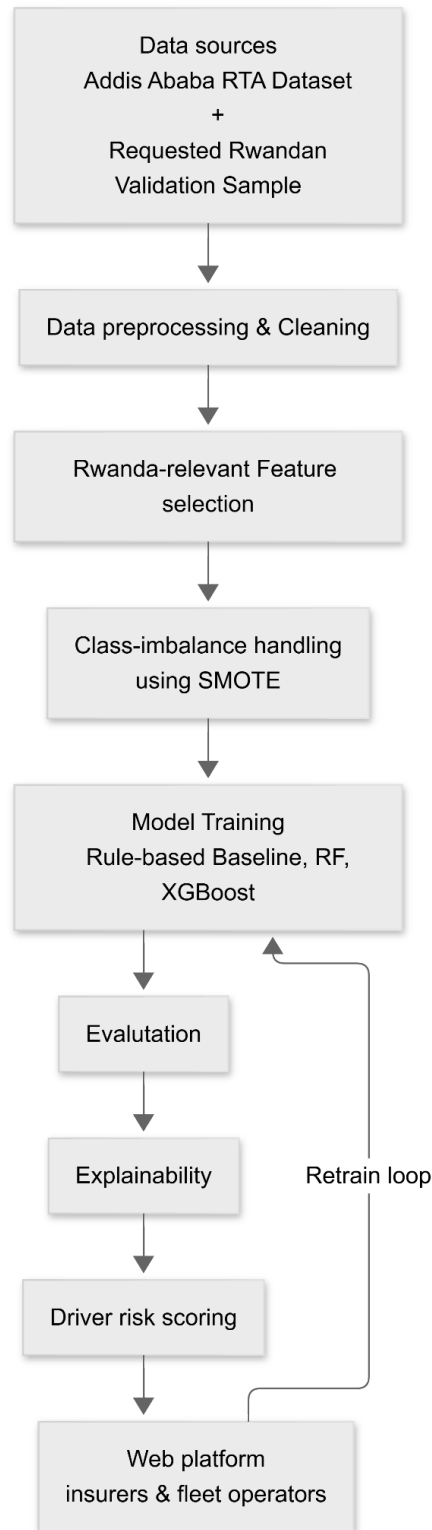
## **CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN**

### **3.1 Introduction**

This chapter describes the methodology and design of the MBERE ML system. The work follows a Design Science Research orientation building and evaluating an artefact that solves a real problem combined with an iterative, incremental development model so that the data pipeline, machine-learning models and web platform can be built and tested in short, verifiable cycles. The research is primarily experimental: data are prepared and engineered, models are trained and compared against measurable metrics, and the resulting predictions are delivered through a prototype platform for the intended beneficiaries.

### **3.2 Research Design and Proposed Model**

The proposed methodology is summarized in Figure 1. Real crash-record data from the Addis Ababa Road Traffic Accident dataset (Bedane, 2020), together with a requested Rwandan validation sample, are cleaned and preprocessed. Features relevant to the Rwandan road environment are then selected and engineered including driver age, sex and experience, vehicle type (with motorcycle treated as a first-class category), road surface, light condition, weather and time of day. Because crash outcomes are imbalanced, the Synthetic Minority Over-sampling Technique (SMOTE) is applied before training. A transparent rule-based baseline is then compared against Random Forest and XGBoost models, which are evaluated using precision, recall, F1-score and AUC under stratified k-fold cross-validation. SHAP is used to explain the predictions, the model emits a driver risk score and band, and the result is served through a web platform. A retraining loop returns new labelled outcomes to the training stage so the model can be periodically updated as Rwandan validation data become available.



*Figure 1: Proposed MBERE ML methodology and machine-learning pipeline.*

### 3.3 Class Diagram

Figure 3 shows the principal classes of the platform and their relationships. The Driver class holds driver attributes; Feature Record captures the engineered driver-and-context features used for prediction; Risk Model encapsulates training, prediction and explanation; and Risk Assessment records a driver's computed risk score, band and the model version used. The User and Dashboard classes represent the insurer or fleet operator and the interface through which they view and analyse risk assessments.

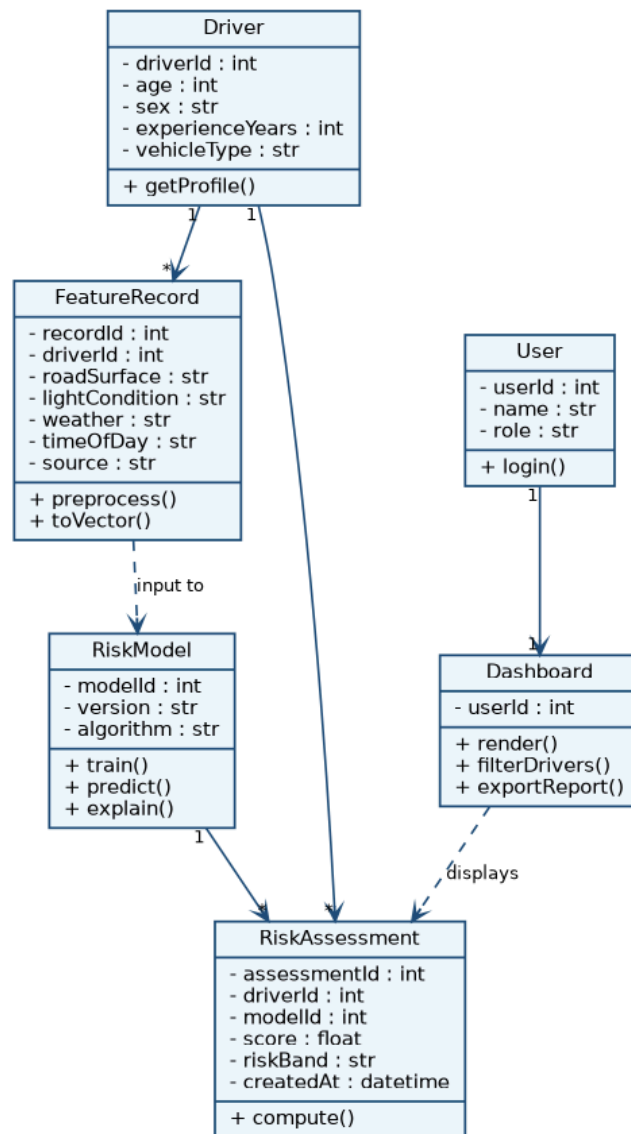


Figure 3: System class diagram.

### 3.4 System Architecture

The architecture, shown in Figure 2, is organized into four layers. The data layer stores the crash-record dataset and the Rwandan validation sample, together with a feature store of engineered features. The machine-learning layer holds the trained model (Random Forest or XGBoost), an inference API that returns risk scores, and a SHAP explainer. The application layer is a web dashboard presenting a driver list, individual risk scores, per-driver explanations and a portfolio view. The users are motor insurers and fleet or motorcycle-taxi operators.

A retraining pipeline feeds new labelled outcomes from the application back into the feature store and model, supporting periodic updates.

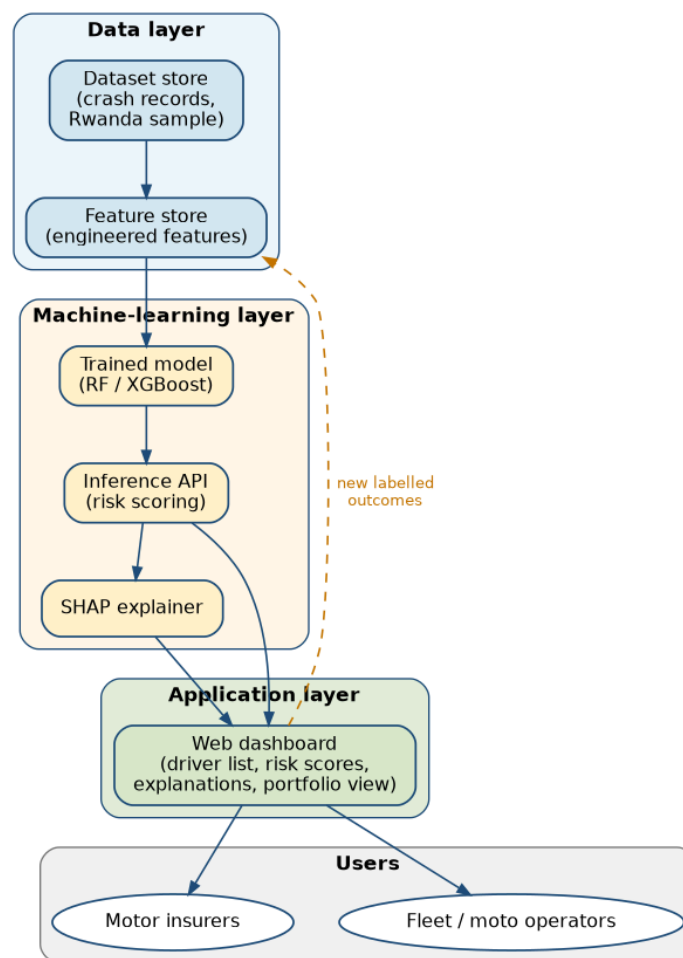


Figure 2: MBERE ML system architecture.

### 3.5 Entity-Relationship Diagram

The entity-relationship diagram in Figure 4 models the data the platform persists. A DRIVER has many FEATURE\_RECORD entries and receives many RISK\_ASSESSMENT records over time. Each MODEL\_VERSION produces many risk assessments, allowing predictions to be traced to the model that generated them, and each APP\_USER (an insurer or fleet operator) reviews many risk assessments. This structure supports auditability, versioning and the retraining loop described above.

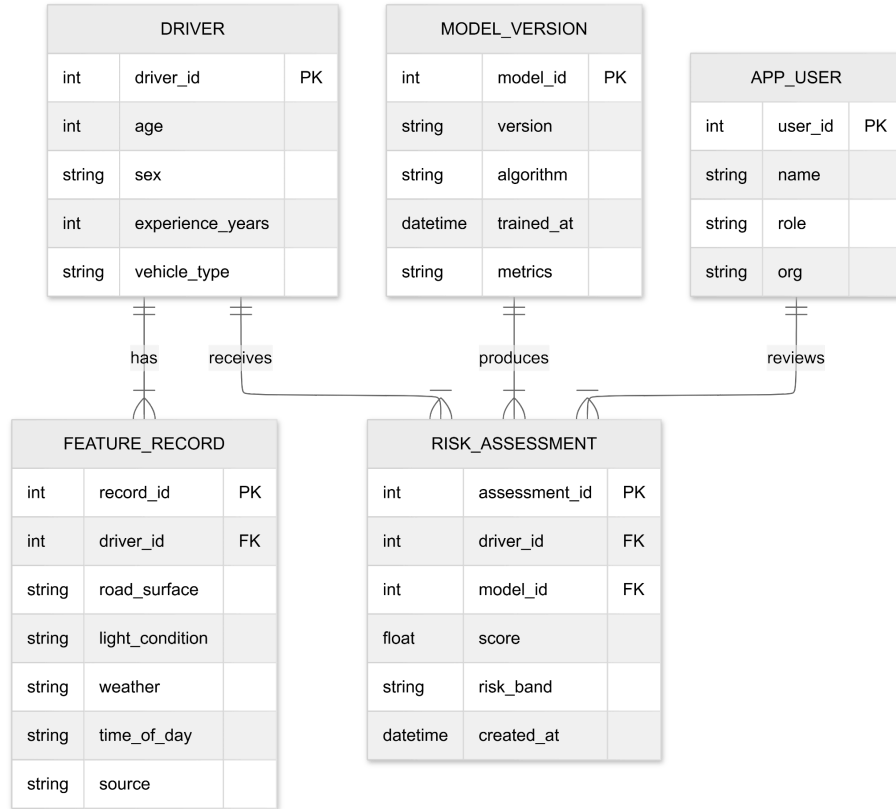


Figure 4: Entity-relationship diagram (ERD).

### 3.6 Development Tools

- Data and machine learning: Python with pandas and NumPy for data preparation, scikit-learn for the rule-based baseline and Random Forest, the XGBoost library for gradient boosting, imbalanced-learn for SMOTE, and SHAP for explainability.
- Datasets: the Addis Ababa Road Traffic Accident dataset (Bedane, 2020) as the primary source, the Porto Seguro (2017) Safe Driver Prediction dataset as a driver-level benchmark, and a requested Rwandan validation sample from the Rwanda National Police and the Rwanda Biomedical Centre.

- Back-end and data storage: a REST API built with Flask or FastAPI and a relational database such as PostgreSQL for drivers, feature records, model versions and risk assessments.
- Web platform: a lightweight front-end (React) presenting the driver list, risk scores, SHAP-based explanations and portfolio view.
- Environment and tooling: Jupyter notebooks or Google Colab for experimentation, Git and GitHub for version control, and Matplotlib and Graphviz for analysis figures and diagrams.

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